International Islamic University Chittagong
Department of Computer Science and Engineering
Program: B.Sc. in CSE: Semester: 5th
Mid-Term Examination, Spring-2025

Course Code: CSE 3523 Course Title: Microprocessor, Microcontroller and Embedded System
Time: 1 hour 30 minutes Full Marks: 30

Answer the following Three (3) questions. Each question carries 10 marks.

Questions

1.	a)	Describe the physical address calculation in the 8086 microprocessor. Include how segment and offset registers are used. Find the physical address of 1000:2000.	CO1	2
1.	b)	Write short notes on the following: i) CISC ii) RISC iii) ALU	CO1	3
1.	c)	Explain in detail the steps involved in the execution cycle of a microprocessor.	CO2	5
2.	a)	Draw the internal block diagram of the 8086 microprocessor.	CO1	4
2.	b)	what are the addressing modes used in the following Instructions? If the instructions are incorrect, then explain the reasons and make them correct **i.MOV AL, BL** **ii.MOV AL, BX** **iii.MOV AH, [BX]** **iv. MOV CL, 3A8H**	CO1	2
2.	c)	Write a short assembly program to add two 16-bit hexadecimal numbers 20F1H and 3C72H using 8-bit addition. Also, explain how the ADC instruction works in this context.	CO2	4

OR,

2.	c)	**Find the value of** C, A, P, Z, S, O FLAGs after execution of this instruction. **MOV AL, 3FH** **MOV BL, 2AH** **ADD AL, BL**	CO1	4

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3.	a)	Write about the advantages and disadvantages of low-level language.	CO2	2
3.	b)	Write an assembly language program to set the rightmost 3-bits, to clear the leftmost 2-bits, and to Invert the bit position 5, 7 & 8th of register value AX=FF0EH.	CO2	4
3.	c)	Given, AX=8877 H and BX=79ABH. Interchange the value of AX and BX using PUSH and POP Operation. Draw the Table of AX, BX and SP Value after every instruction. Take the initial address, SP= 2009H.	CO2	4

OR

3.	c)	What is the operation of the registers in the 8086 microprocessor during 8-bit and 16-bit multiplication? Write assembly language for the equation y=6×8×10H.	CO2	4

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Department of Computer Science and Engineering

Midterm Examination, Spring- 2024

Course Code: CSE-3523, Course Title: Microprocessors, Microcontrollers and Embedded Systems

Total Marks: 30 Time: 1.5 Hours

(There are Three questions, answer all Three Questions. Figures on the right margin indicate marks.)

1.	**a) What are the basic components of a microprocessor? What makes the microprocessor different from the CPU? Which has more efficiency, and how?** **OR,** **Consider a scenario in which a library checkout system operates as an analogy for system buses. In this analogy, the book reader will be borrowed book, books are keep in the bookshelf, and the librarian will provide book to the book reader. Explain how these components interact in the library system, and provide insights into the roles of the system bus in facilitating the borrowing process. Additionally, highlight how the return of a book can be correlated to the system bus analogy.**	**CO1**	**3**
	b) Consider the following instructions: **1. MOV BL, 0032H** **2. SUB CS, BX** **3. MOV CX, 0041H** **Are the above instructions correct or incorrect? Explain if the instructions are right or wrong with logic.**	**CO2**	**3**
	c) How does the fetch-decode-execute cycle work? Describe for instruction MOV AX,0x100. How can the efficiency of the Fetch-Decode-Execution cycle be affected by the choice of instruction format and addressing modes?	**CO3**	**4**

2.	***a) Draw the Internal Block Diagram of 8086 Microprocessor.*** **OR,** ***a) What is memory segmentation for 8086 processor? What is the use of it? How can a 16-bit microprocessor generate 20-bit memory addresses?***	**CO2**	**3**
	b) Consider the following instruction:MOV CH,[BX][SI]+2234H. Where CS=0200H,IP=0000H,BX=2000H,SI=3000H. The memory content of the first 4 byte instruction is 8AA03412H.Calculate the physical address(before and after)passing the instruction. Which memory addressing mode will the instruction use?Explain with the help of diagram.	**CO2**	**4**
	C) In the real mode, show the starting and ending addresses of each segment located by the following segment register values: **I. DA01H** **II. 23F1H** capacity = End-start	**CO2**	**3**

3.	**a) What is instruction? What is the main difference between an opcode and an operand in a microprocessor? Explain with examples.**	**CO3**	**3**
	b)Consider the following scenario: **1. Tanveer and Shihab are good friends. Tanveer was preparing himself for the midterm exam, sitting in the library. At that time, Shihab called Tanveer and asked him to come and play at Central Field. Tanveer avoided him and said he would come after 30 minutes after completing his**	**CO2**	**4**

assignment. At the same time, Tanveer's teacher called and asked to meet to arrange a special class. Tanveer went to meet without avoiding. Shihab noticed something in the pin diagram of the microprocessor (intel 8086): the timing of the interrupt was also the same. Some interrupts are avoided by the microprocessor, while some are not.

Now write and discuss the pins responsible for the interrupts.

2. DMA stands for Direct Memory Access. It is designed by Intel to transfer data at the fastest rate. It allows the device to transfer the data directly to/from memory without any interference of the CPU.

Now, discuss the pins responsible for DMA techniques in 8086 μp.

OR,

In an arithmetic operation involving the 8086 Microprocessor, describe how the general-purpose registers AX, BX, CX, and DX are utilized. Discuss the specific roles each register plays during the execution of the operation. Provide an example arithmetic operation and illustrate the step-by-step involvement of these registers in the process.

c) Write an assembly language program to compute Y= (5+3) * (2+1)	CO2	3

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Department of Computer Science and Engineering

B.Sc in CSE Mid Examination, Autumn 2023

Course Code: CSE-3523

Course Title: Microprocessors, Microcontrollers and Embedded Systems

Time: 1 hour 30 minutes **Total Marks:** 30

- Answer all **Three** of the following questions.
- The figures in the right-hand margin indicate full marks.

1 a)	What are the essential components of a microprocessor? What makes the microprocessor different from the microcontroller?	3	CLO 1 U
b)	Suppose you need to run a high-quality and speedy performance video on computers. What type of bus you will use to support this operation on the computer? Describe the bus.	3	CLO 1 AP
c)	You are sitting at a specific place inside a bus. The ticket checker asks you to show your ticket. Do you see any similarity between address, control, and data bus with the above-mentioned scenario? If yes, explain with logic.	4	CLO 1 AP
2 a)	In the microprocessor, what is the function of IP and ALU?	2	CLO 1 R
or,	What determines whether a microprocessor is considered an 8-bit, 16-bit, or 32-bit device?		
b)	What is memory segmentation for the 8086 processor? What is the use of it? How can a 16-bit microprocessor generate 20-bit memory addresses?	4	CLO 1 R
or,	Draw the Internal Block Diagram of the 8086 Microprocessor.		
c)	If the stack segment register contains 56C0H and the stack pointer register contains 1358H, what is the physical address of the top of the stack? 57F58H	2	CLO 1 AP
d)	The value of the Code Segment (CS) Register is 4042H and the value of different offsets is as follows: BX: 2025H, IP: 0580H, DI: 4247H Calculate the effective address of the memory location pointed by the CS register. 409A0H	2	

3 a) Differentiate between Mnemonic and Pseudo-op. 1 CLO 1 U

b) Consider the following instructions: 4 CLO 1 E

A. MOV AX, BL
B. MOV BL, 0032H
C. MOV AX, [BX]
D. MOV AX, CS

What are the addressing modes used in the above statements? Are the above instructions correct or incorrect? Explain if the instructions are right or wrong with the logic.

or, Which of the following are legal numbers or legal names? If the number or name is illegal explain the reason.

I. Numbers:
   A. 1,101B
   B. FFFFH
   C. -55D
   D. 1FH

II. Names:
   A. iandhe
   B. i&he
   C. A45.26
   D. A2

c) Write an assembly language program in Microprocessor 8086 to compute: 5 CLO 2 AP

I. Y= A + B
II. X= 2Y - 3Z

International Islamic University Chittagong

Department of Computer Science and Engineering

B. Sc. in CSE Midterm Examination, Spring 2023

Course Code: CSE 3523 Course Title: Microprocessors, Microcontrollers and Embedded Systems

*Total marks: **30***

*Time: **1** hours **30** minutes*

[Answer all the questions; in some questions, there are options; solve the one which you have been instructed to solve;
Precisely follow the guideline for preparing and submitting the answer script;
Figures in the right hand margin indicate full marks.]

			CO	DL
1.				
a)	Differentiate between CPU and Microprocessor.	2	CO1	C2
b)	You are sitting at a specific place inside a bus. The ticket checker asks you to show your ticket. Do you see any similarity between address, control and data bus with the above mentioned scenario. If yes, explain with logic.	3	CO1	C2
c)	Distinguish between AX and DX register. Write down the functionality of pointer and index register with example.	3	CO1	C1
d)	Suppose, you connect a printer to a computer. What kind of interface do you use for printer? Why? Explain reason for your answer.	2	CO1	C1
2.				
a)	What are the addressing modes used in 8086 microprocessor. Name the addressing modes with example. Differentiate between op code and operands.	3	CO3	C1
b)	Consider the following instructions : 1. MOV AX, BL 2. MOV CL,004H 3. SUB AX,BX What are the addressing modes used in the above statements? Are the above instructions correct or incorrect? Explain if the instructions are right or wrong with logic.	3	CO3	C2

MOV CX [illegible] MOV CX,[AX+BX+3345H]

[AX] ([illegible] your Student Id)H

(BX)= (last four digits of your student id*2)H

Initial value of IP= 0002H

Value of CS= 01000H CS =0100H

N.B: If there are any missing values in the above mentioned instruction, assume the value of the register with appropriate size. Calculate the Physical address both before and after instruction execution and show the internal operations using figure.

3.

a)	What is the difference between instruction fetch, decode and execute? Explain with example.	3	CO1
b)	Write short notes on the following :	6+ 1= 7	CO1

1. MAR
2. MBR
3. PC
4. IR
5. $\overline{CS}$
6. $\overline{R}$/W

What is decoder? Why is it used during instruction fetching and executing?

Sidrati

Sep 26, 2022

International Islamic University Chittagong

Department of Computer Science and Engineering

B.Sc in CSE — Mid Term Examination, Spring-2022

Course Code: CSE-3523

Course Title: Microprocessors, Microcontrollers & Embedded Systems

Time: 1 hour 30 minutes — **Total Marks:** 30

- Answer all **Three** of the following questions.
- The figures in the right-hand margin indicate full marks.

1 a) In the microprocessor, what is the function of *IP* and *ALU* ? 2 (CLO1)

b) Consider a machine language instruction that moves a copy of the contents of register *BX* in the CPU to a memory word. What happens during 4 (CLO1)

I. the fetch cycle?

II. the execution cycle?

c) What is the use of a register in a microprocessor? Briefly describe the registers of Microprocessor 8086. 4 (CLO1)

or, Why is memory divided into segments? Explain according to Microprocessor 8086.

2 a) Draw the internal architecture of the 8086 microprocessor. 4 (CLO1)

b) Calculate the physical address from the following Segment: Offset pairs - 4 (CLO3)

I. 2900H: 3A00H

II. 1239H: 0000A900H

or, If the stack segment register contains 56C0H and the stack pointer register contains 1358H, what is the physical address of the top stack?

c) Find out the memory size in MB when address lines are - 2 (CLO1)

I. 34 bits

II. 128 bits

3 a) What types of programs are usually written in assembly language? 2 (CLO1)

or, What is Opcode? Give an example.

b) Write whether each of the following instructions is **Legal** or **Illegal**. 2 (CLO3)
W1 & W2 = word variables

I. ADD B1, 122
II. MOV DS, 100H
III. XCHG W1, W2
IV. MOV CS, DS

c) Write the types (Decimal, Octal, etc) of the following numbers in assembly language. Put 'x' for the Illegal number(s). 2 (CLO3)

I. 1,101 v
II. 1010 D
III. FFFFH x
IV. -55D D

d) Translate the High-Level Language into Assembly Language. Add proper comments for each instruction. 4 (CLO3)

A = 2B - 3A

CS CamScanner